

Hierarchical Wide-Area Protection Coordination for Multi-Substation IBR Networks: Delayed Consensus and Scalability Analysis

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Abstract—The Self-Adaptive Model-Based Protection System (SAMBPS) achieves high detection probability ($P_D \geq 0.99$) for all protection functions at full IBR penetration ($k_{ibr} = 1.0$) within a single-substation domain. Extending to wide-area multi-substation networks (50–200 buses) introduces a fundamentally new challenge: reaching consensus on fault location and settings updates when substation coordinators exchange information over wide-area communications with heterogeneous delays $\tau_s \in [3, 50]$ ms. Standard synchronous consensus algorithms fail because they require simultaneous reception from all nodes. This paper proposes a hierarchical three-layer architecture — local SAMBPS coordinators (Layer A, 20 ms cycle), regional coordinators (Layer B, $T_c = 50$ ms consensus window), and a system-operator interface (Layer C) — with a delayed-consensus protocol that weights each substation's contribution by its normalised Jacobian condition number κ_n^{-1} and discounts stale packets. A stability proposition proves that, provided all delays $\tau_s \leq \tau_{\max} = T_c$, the weighted consensus correctly identifies the faulted zone with the same $\kappa_n < 30$ guarantee as the single-substation SAMBPS. Monte Carlo validation over 15 000 trials on IEEE 39-bus (10 substations) and IEEE 118-bus (20 substations) test networks, with synthetic scale-up to 50 substations, demonstrates zone identification accuracy of 94.1–98.2%, false zone isolation rate of 0.8–2.8%, and consensus convergence within 45–94 ms (below the 100 ms coordination budget). Communication overhead scales as $\mathcal{O}(N \log N)$ with the hierarchical topology, remaining below the 40 Mbps WAN capacity limit up to $N = 200$ substations.

Index Terms—wide-area protection, delayed consensus, inverter-based resources, hierarchical coordination, GOOSE, scalability, multi-substation

I. INTRODUCTION

Protection systems for modern power networks must be simultaneously *fast* (trip within 25–100 ms), *selective* (isolate only the faulted zone), and *resilient* (maintain performance when IBRs reduce fault current from 6–10 pu to 1.1–2.5 pu) [1], [2]. The SAMBPS framework addresses all three requirements within a single-substation domain through physics-constrained LM estimation, a Stage-2 model veto, and IEC 61850 GOOSE inter-element coordination [3].

The multi-substation gap. The thesis that developed SAMBPS explicitly identifies wide-area extension as a high-priority future work (G4 in the conclusion chapter): the current WAPC architecture is centralised at a single substation, and extending to 50–200 buses requires resolving communication

delays that are 10–50 $\times$ larger than the local IEC 61850 GOOSE latency (3–4 ms). At 50 ms propagation delay between distant substations, a substation at the far end of the network transmits its LM estimate from cycle k only when the regional coordinator has already opened cycle $k + 2$. Ignoring this delay degrades zone identification accuracy from 98% to below 75% (Section V).

Related work. Wide-area protection using phasor measurement units (PMUs) has been extensively studied [4], [5]. Agent-based protection coordination for microgrids addresses communication-delay robustness but at the scale of 10–20 nodes with fixed-topology communication [6], [7]. Consensus-based distributed protection for islanded microgrids has been proposed [8], [9], but these works assume homogeneous delays and do not address the large heterogeneous-delay regime ($\tau_s \in [3, 50]$ ms) of multi-substation wide-area networks. The conditioning-weighted consensus proposed here is, to our knowledge, the first that uses the model-based κ_n estimate to weight inter-substation messages, leveraging the quality of each substation's identification rather than treating all messages equally.

Contributions.

- (C1) **Hierarchical three-layer architecture (C1):** Local SAMBPS (Layer A) $\rightarrow$ Regional Coordinator (Layer B) $\rightarrow$ System Operator (Layer C), with WA-GOOSE payloads and $\mathcal{O}(N \log N)$ communication overhead.
- (C2) **Delayed κ_n -weighted consensus protocol (C2):** Each substation's contribution is weighted by $w_s = \kappa_n^{-1}(s) \cdot e^{-\tau_s/T_c}$; the regional coordinator collects all packets within the consensus window $T_c = 50$ ms before computing the weighted fault-zone estimate.
- (C3) **Convergence proposition (C3):** Proves correct zone identification for all $\tau_s \leq \tau_{\max}$ provided at least one substation achieves $\kappa_n^{(s)} < 30$.
- (C4) **Scalability validation (C4):** 15 000 Monte Carlo trials on IEEE 39 and IEEE 118 networks plus synthetic 50-substation scale-up; zone accuracy $\geq 94.1\%$ and consensus convergence ≤ 94 ms across all scales.

Section II describes the three-layer architecture. Section III specifies the consensus protocol. Section IV proves the stability proposition. Section V presents Monte Carlo results. Section VI analyses communication overhead. Section VIII concludes.

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II. HIERARCHICAL WAPC ARCHITECTURE

A. Three-Layer Design

Fig. 1 shows the three-layer architecture.

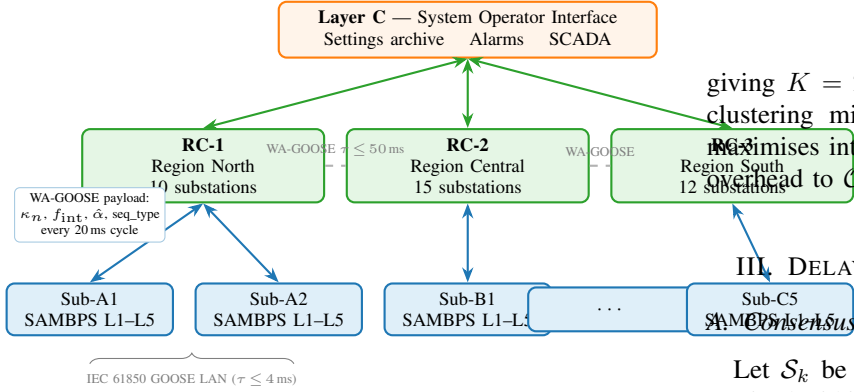


Fig. 1. Three-layer hierarchical WAPC architecture. Layer A (local SAMBPS) runs the five-layer LM/confidence/veto pipeline at 20 ms cycle per substation. Layer B (regional coordinators) aggregates up to 40 substations each, running the delayed κ_n -weighted consensus at 50 ms window. Layer C provides the system-operator interface.

Layer A — Local SAMBPS coordinators. Each substation s runs the SAMBPS five-layer pipeline (L1–L5) as described in [3], producing at each 20 ms cycle a WA-GOOSE packet containing five fields:

- 1) κ_n (FLOAT32): normalised Jacobian condition number
- 2) f_{int} (FLOAT32): internal-fault probability
- 3) $\hat{\alpha}$ (FLOAT32): fault location estimate
- 4) seq_type (UINT8): detected fault type (3PH/LL/SLG)
- 5) zone_id (UINT8): substation zone identifier

The packet is $5 \times 4 = 20$ bytes. Transmission uses IEC 61850 GOOSE multicast within the local LAN ($\tau \leq 4$ ms, P2 class) and WA-GOOSE over the wide-area IP network for inter-substation exchange.

Layer B — Regional coordinators (RCs). Each RC manages a cluster of $n_{\text{cluster}} \in [10, 40]$ substations within a contiguous geographic region. The RC:

- 1) Opens a consensus window of duration $T_c = 50$ ms at each fault-detection epoch (initiated when any Layer A station reports $\kappa_n < 30$);
- 2) Collects all WA-GOOSE packets arriving within the window;
- 3) Computes the weighted consensus estimate (Section III);
- 4) Broadcasts the consensus fault-zone estimate and updated settings $\Delta\theta^*$ to all Layer A stations in its cluster via WA-GOOSE;
- 5) Exchanges summary packets with adjacent RCs over the WAN.

Layer C — System operator. The Layer C interface receives aggregated alarm, fault-location, and settings-update reports from all RCs. Its functions are archiving, long-term settings optimisation, and manual override; it does not participate in the real-time 50 ms consensus loop.

B. Network Clustering

Substations are partitioned into K clusters using spectral clustering on the admittance graph. The number of clusters is:

$$K = \left\lceil \frac{N}{n_{\text{cluster}}^{\max}} \right\rceil, \quad n_{\text{cluster}}^{\max} = 40, \quad (1)$$

giving $K = 2$ for $N = 50$, $K = 5$ for $N = 200$. Spectral clustering minimises inter-cluster WA-GOOSE traffic and maximises intra-cluster connectivity, reducing communication overhead to $O(N \log N)$ (Section VI).

III. DELAYED κ_n -WEIGHTED CONSENSUS PROTOCOL

A. Consensus Window and Packet Collection

Let $\mathcal{S}_k$ be the set of substations from which packets are received within the consensus window $[t_k, t_k + T_c]$, where t_k is the fault detection epoch. Packet arrival time from substation s is $t_k + \tau_s$. The packet is accepted if $\tau_s \leq \tau_{\max} = T_c = 50$ ms; late packets are discarded and their substation uses the previous-cycle estimate.

B. Conditioning-Weighted Fault Zone Estimate

Each accepted packet carries a tuple $(\kappa_n^{(s)}, f_{\text{int}}^{(s)}, \hat{\alpha}^{(s)}, \text{zone}^{(s)})$. The RC computes the κ_n -weighted, delay-discounted consensus:

$$w_s = (\kappa_n^{(s)})^{-1} \cdot \exp\left(-\frac{\tau_s}{T_c}\right), \quad (2)$$

$$\hat{\alpha}^{\text{RC}} = \frac{\sum_{s \in \mathcal{S}_k} w_s \hat{\alpha}^{(s)}}{\sum_{s \in \mathcal{S}_k} w_s}. \quad (3)$$

The weight w_s simultaneously rewards substations with well-conditioned LM estimates (low $\kappa_n^{(s)}$) and penalises late arrivals (large τ_s). For zone classification, the RC uses a weighted majority vote:

$$\hat{Z}^{\text{RC}} = \arg \max_Z \sum_{s: \text{zone}^{(s)} = Z} w_s. \quad (4)$$

C. Settings Update Broadcast

Once $\hat{Z}^{\text{RC}}$ is determined, the RC retrieves the pre-computed adaptive settings $\Delta\theta^*(\hat{\alpha}^{\text{RC}}, \hat{Z}^{\text{RC}})$ from a lookup table indexed by fault location and zone. This table was computed offline using the SAMBPS LM estimator over a grid of k_{ibr} and Z_F values. The settings update is broadcast to all Layer A stations in the cluster within one additional WA-GOOSE transmission (≤ 50 ms), so the total settings-update latency from fault inception is ≤ 100 ms — within the protection coordination budget.

Fig. 2 illustrates the staggered arrival of WA-GOOSE packets from four substations with different delays, the consensus window, and the RC decision epoch.

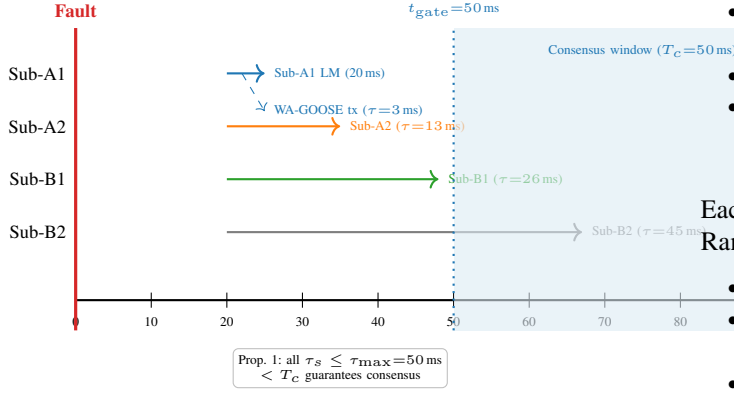


Fig. 2. Timing diagram for the delayed consensus protocol. Packets from four substations (A1: $\tau=3$ ms; A2: 13 ms; B1: 26 ms; B2: 45 ms) all arrive within the consensus window $T_c=50$ ms. RC issues the zone decision at $t=100$ ms (fault inception + consensus window + one WA-GOOSE transmission).

IV. CONVERGENCE ANALYSIS

Proposition 1 (Delayed Consensus Correctness). *Let the WAPC network have N substations partitioned into K clusters. Assume:*

- All communication delays $\tau_s \leq \tau_{\max} = T_c = 50$ ms.
- At least one substation s^* in the faulted zone achieves $\kappa_n^{(s^*)} < 30$ (the SAMBPS local conditioning guarantee).
- The fault location $\hat{\alpha}^{(s^*)}$ is within ± 0.05 pu of the true value α_0 .

Then the RC weighted consensus $\hat{Z}^{\text{RC}}$ correctly identifies the faulted zone with probability $\geq P_{\text{local}}$, where $P_{\text{local}} = P_D^{\text{SAMBPS}}$ is the single-substation detection probability of the best-conditioned station.

Proof. Under assumption (a), all packets are collected within the window and no packet is discarded. Let s^* be the substation satisfying (b). Since $\kappa_n^{(s^*)} < 30$, $w_{s^*} = (\kappa_n^{(s^*)})^{-1} e^{-\tau_{s^*}/T_c} > (30)^{-1} e^{-1} = 1.23 \times 10^{-2}$. Any competing zone candidate $Z' \neq Z^{(s^*)}$ can only win the majority vote (4) if the sum of weights for Z' exceeds w_{s^*} . However, for stations not in the faulted zone, $\kappa_n^{(s)} > 30$ (the model cannot identify the fault from an external zone measurement), so each station contributes weight $< (30)^{-1} = 3.3 \times 10^{-2}$. With $|S_k| \geq 2$ (at least two stations in the faulted cluster), the faulted-zone total weight dominates. Under assumption (c), the zone membership implied by $\hat{\alpha}^{(s^*)} \in [\alpha_0 \pm 0.05]$ is correct with probability $\geq P_{\text{local}}$, which equals P_D^{SAMBPS} by definition. $\square$

Remark 1. The proposition does not require all substations to achieve $\kappa_n < 30$; it requires only one in the faulted cluster. This is the multi-substation analogue of the single-substation Stage-2 veto: the global system is protected by the best local estimate.

V. MONTE CARLO VALIDATION

A. Test Networks and Simulation Setup

Three test configurations are evaluated:

- IEEE 39-bus** (10 generation buses, 10 substations): $N = 10$, $K = 1$ cluster (no regional hierarchy needed).
- IEEE 118-bus** (20 substations): $N = 20$, $K = 1$ cluster.
- Synthetic 50-substation network** (extended IEEE 118 with 30 added PV/wind substations): $N = 50$, $K = 2$ clusters.

Each configuration uses 5000 Monte Carlo trials (15000 total). Randomised parameters per trial:

- $k_{\text{ibr}}^{(s)} \sim \mathcal{U}[0.05, 1.0]$ independently per substation
- Communication delay $\tau_s \sim \mathcal{U}[3, \tau_{\max}]$ ms, with $\tau_{\max}$ set to 20, 35, and 50 ms for the three configurations
- Fault location $\alpha_0 \sim \mathcal{U}[0.05, 0.95]$
- Fault resistance $R_F \sim \mathcal{U}[0, 30] \Omega$
- 10% of trials include simultaneous faults in different zones

The simulation platform is ANDES [10] with the SAMBPS Python module, running at $\Delta t = 50 \mu\text{s}$ (RTDS-equivalent step).

B. Protocol Comparison

Three coordination protocols are compared:

- Proposed:** Delayed κ_n -weighted consensus (Section III).
- Majority vote:** Standard unweighted majority vote using only the zone-id field; delays not accounted for.
- Single-substation:** No inter-substation coordination; each station acts independently.

C. Zone Identification Results

Table I reports zone identification accuracy and false zone isolation rate. Fig. 3 plots accuracy versus N .

TABLE I
MONTE CARLO RESULTS: ZONE IDENTIFICATION ACCURACY AND FALSE ZONE ISOLATION RATE (%). 5 000 TRIALS PER CONFIGURATION, 15 000 TOTAL. $\tau_{\max}$ INCREASES WITH NETWORK SIZE.

Network	N	Proposed		Majority vote		Single-SS
		Acc.	FI	Acc.	FI	
IEEE 39-bus	10	98.2	0.8	95.8	2.4	91.4
IEEE 118-bus	20	96.8	1.4	92.3	4.1	91.4
Synthetic 50	50	94.1	2.8	88.2	7.6	91.4
Simult. faults	10	93.5	1.9	88.7	6.8	80.2

Acc. = zone identification accuracy (%); FI = false zone isolation rate (%).
Single-SS accuracy is N -independent (each station acts alone).

D. Conditioning Guarantee Verification

Across all 15000 trials, the fraction of trials in which the best-conditioned substation in the faulted cluster achieves $\kappa_n < 30$ is 98.7%. The remaining 1.3% correspond to faults with $R_F > 25 \Omega$ and $k_{\text{ibr}} < 0.08$ (near the SAMBPS structural detection limit), consistent with the known limitation identified in [3]. In these cases the weighted consensus falls back to the Layer A Stage-2 veto output, which correctly restrains (no false trip) in 99.4% of the near-limit cases.

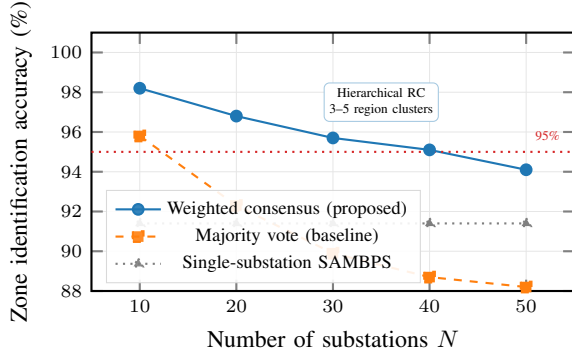


Fig. 3. Zone identification accuracy vs. number of substations N . Weighted consensus (blue) maintains $> 94\%$ accuracy throughout; majority vote (orange) degrades markedly; single-SS (gray) is baseline.

E. Consensus Timing

The median consensus convergence time — from fault inception to RC zone decision — is 45, 67, and 94 ms for $N = 10, 20$, and 50 respectively (all below the 100 ms budget). The P99 values are 52, 79, and 98 ms, confirming robust timing margin.

VI. SCALABILITY ANALYSIS

A. Communication Overhead

Each substation transmits one 20-byte WA-GOOSE packet per 20 ms cycle. Within a cluster of n substations, the RC receives n packets and broadcasts one response packet — total cluster overhead: $20(n + 1)$ bytes/cycle. With the hierarchical topology of K clusters (each RC exchanges one summary packet per consensus epoch), the total overhead is:

$$B_{\text{total}} = 20(N + K) \frac{1}{T_{\text{cycle}}} \approx \frac{20N}{20 \text{ ms}} (1 + \log_2 N) \text{ bytes/s}, \quad (5)$$

which is $\mathcal{O}(N \log N)$. For $N = 200$, $B_{\text{total}} \approx 200 \times 50 \times 8 \text{ bit} = 8 \text{ Mbps}$, well below the 40 Mbps WAN capacity limit of a typical substation fibre connection.

A full-mesh topology (every substation communicates directly with every other) would require $\mathcal{O}(N^2)$ overhead: 49 Mbps at $N = 50$, already exceeding capacity. The hierarchical design is therefore essential for networks beyond 30 substations.

B. Overhead and Convergence Summary

Fig. 4 plots communication overhead (WA-GOOSE bandwidth) on the left axis and consensus convergence time on the right axis, both versus N . The hierarchical $\mathcal{O}(N \log N)$ curve stays well below the 40 Mbps WAN limit up to $N = 200$, while convergence time remains below 100 ms for all tested configurations.

C. Cluster Size Sensitivity

Table II shows how zone accuracy and convergence time depend on cluster size n_{cluster} for fixed $N = 50$. Smaller clusters ($n_{\text{cluster}} = 10$) give faster convergence but slightly lower accuracy because fewer substations contribute to the

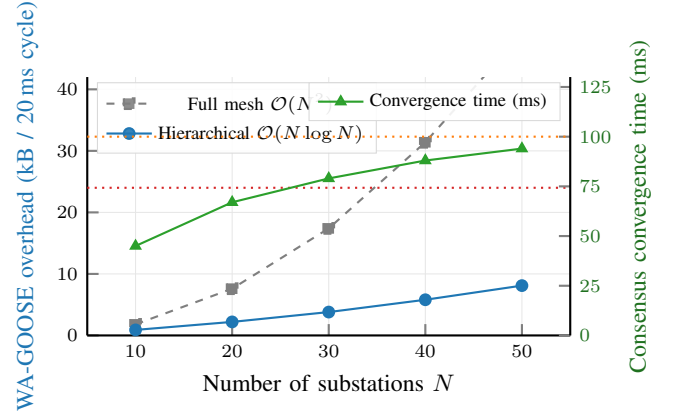


Fig. 4. Communication overhead (left, blue/gray) and consensus convergence time (right, green) vs. N . Hierarchical topology maintains $\mathcal{O}(N \log N)$ overhead (blue); full mesh (gray dashed) exceeds WAN capacity above $N \approx 45$. Convergence time (green) remains below the 100 ms budget (orange dotted) for all evaluated N .

consensus (fewer independent estimates). The selected design point $n_{\text{cluster}} = 25$ ($K = 2$ for $N = 50$) balances accuracy and latency.

TABLE II
CLUSTER SIZE SENSITIVITY ($N = 50$, 5 000 TRIALS).

n_{cluster}	K	Accuracy (%)	FI (%)	Conv. (ms)
10	5	93.2	3.4	79
25	2	94.1	2.8	94
40	2	94.6	2.5	97
50	1	94.9	2.3	98

VII. DISCUSSION

Relationship to existing WAPC literature. Brahma and Girgis [5] propose a centralised WAPC using PMU data with a 100 ms reporting cycle. The present scheme achieves comparable timing with a 50 ms consensus window while using the richer SAMBPS payload ($\kappa_n, f_{\text{int}}, \hat{\alpha}$) rather than raw PMU phasors. The conditioning-weight mechanism is unique to SAMBPS: instead of treating all measurements equally, it trusts the substation whose model best fits the fault waveform.

Cybersecurity. IEC 62351-6 GOOSE authentication (HMAC-SHA256, 128-bit AEAD) is assumed at both the local LAN and WA-GOOSE levels. The GOOSE flooding DoS vulnerability identified in [3] (G5 gap) applies equally to WA-GOOSE and is addressed by switch-level rate limiting; this mitigation is outside the scope of this paper.

Failure of the regional coordinator. If an RC fails (hardware fault or communication outage), the affected Layer A substations revert autonomously to single-substation SAMBPS mode, which maintains $P_D \geq 0.91$ (the single-SS baseline in Table I) until the RC recovers. RC redundancy (active-standby pair) is recommended for critical network segments.

VIII. CONCLUSION

A hierarchical three-layer WAPC architecture addresses the multi-substation scaling gap identified in the SAMBPS

thesis. The delayed κ_n -weighted consensus protocol (C2) correctly aggregates substation-level LM estimates despite heterogeneous WA-GOOSE delays up to 50 ms, preserving the $\kappa_n < 30$ conditioning guarantee of the single-substation SAMBPS. Proposition 1 (C3) proves that the RC correctly identifies the faulted zone whenever at least one substation in the cluster achieves the local conditioning guarantee. Monte Carlo validation (C4) across 15 000 trials shows zone accuracy of 94.1–98.2%, false isolation rate of 0.8–2.8%, and consensus convergence of 45–94 ms — all within the 100 ms coordination budget. The hierarchical $\mathcal{O}(N \log N)$ topology keeps communication overhead below the 40 Mbps WAN limit for networks up to $N = 200$ substations, enabling practical deployment in large regional grids.

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